MTH 132(C and K) Dr. James Ashe
Worksheet 5.1 and 5.2 Tuesday 18^{th} March, 2014 **Name:**

Formulas 5.1

P =principal/present value	r =annual interest rate	m =compoundings per year	$n = tm$
A =future/maturity value	t =number of years	$i = r/m$ =rate per period	r_E =effective rate

Simple Interest	Compound Interest	Continuous Compounding	
$A = P(1+rt)$	$A = P(1+i)^n$	$A = Pe^{rt}$	Future
$P = \dfrac{A}{1+rt}$	$P = A(1+i)^{-n}$	$P = Ae^{-rt}$	Present
	$r_E = \left(1+\dfrac{r}{m}\right)^m - 1$	$r_E = e^r - 1$	Effective

Formulas 5.1

S =future value	n =number of periods
R =periodic payment	i =interest rate per period

Future Value of an Ordinary Annuity	Sinking Fund Payment
$S = R\left[\dfrac{(1+i)^n - 1}{i}\right]$	$R = \dfrac{S \cdot i}{(1+i)^n - 1}$

Find the simple interest AND the future value of the loan.

1. \$1500 at 10% for 10 months.

2. \$3000 at 8% for 9 months.

3. \$47,400 at 15.75% for 157 days. Assume 360 days in a year and 30 days in a month.

Find the compound amount for the given deposit and the amount of interest earned.

4. \$1100 at 7.3% compounded semiannually for 10 years.

5. \$1100 at 7.3% compounded monthly for 52 months.

6. \$1100 at 7.3% compounded continuously for 4 years.

Find the amount that should be invested now to accumulate the following amount, if the money is compounded as indicated.

7. \$5500 at 15.75% compounded monthly for 39 months.

8. \$6000 at 3.5% compounded quarterly for 3 years.

9. \$6000 at 3.5% compounded continuously for 3 years.

Solve

10. A company will need \$100,000 in 9 years for a new building. To meet this goal the CFO will invest money into an account paying 9% annual interest rate compounded quarterly. How much money needs to be invested into the account?

11. You win the lottery. You are offered \$13,00 now or \$20,200 in 5 years. If the money can invested at 5.5% compounded annually, which prize will be worth more in 5 years?

Find the future value of each ordinary annuity, if payments are made and interest is compounded as given. Then determine how much of this value is from interest.

12. $R = \$9,200$ (payments); 10% interest compounded semiannually for 7 years.

13. Payments of \$800 into an account paying 6.51% interest compounded quarterly for 12 years.

Determine the interest rate needed to accumulate the following amounts in a sinking fund, with payments as given.

14. Accumulate \$56,000, monthly payments of \$300 over 12 years.

15. Accumulate \$120,000, monthly payments of \$500 over 15 years.

Find the periodic payment that will amount to each given sum under the given conditions.

16. $S = \$100,000$; interest is 5% compounded annually; payments are made at the end of each year for 12 years.

17. $S = \$200,000$; interest is 5% compounded annually; payments are made at the end of each year for 12 years.